

Module 5

Valuing Bonds and Stocks

FINE 3010, Financial Management
Prof. Renping Li, Tulane University

From Module 4 to Module 5

- **Module 4**, the time value of money, the present value of future cash flows

From Module 4 to Module 5

- **Module 4**, the time value of money, the present value of future cash flows
- **Module 5**, the same time-value-of-money principles value bonds and stocks
 - Bond, the present value of its coupons and face value
 - Stock, the present value of its expected dividends

Module 5, Part 1

Bonds

Motivating Example, Silicon Valley Bank



WSJ, March 13, 2023 (2:54), [“The Silicon Valley Bank Collapse, Explained”](#)

Common Types of Corporate Financing

- **Debt**, lending to the company
 - Main forms, bonds and loans
 - Cash flow, a fixed promised amount
 - “Senior”, paid back before equity in bankruptcy
 - No voting rights or control (unless in bankruptcy)

Common Types of Corporate Financing

- **Debt**, lending to the company
 - Main forms, bonds and loans
 - Cash flow, a fixed promised amount
 - “Senior”, paid back before equity in bankruptcy
 - No voting rights or control (unless in bankruptcy)
- **Equity**, stock ownership in the company
 - Cash flow, dividends (and buybacks)
 - “Residual claimant”, paid whatever is left after all promised payments, no guaranteed payout
 - Voting rights

Common Types of Corporate Financing

	Debt	Equity
Payment	Fixed promised payment	Residual (whatever is left over)
Priority in bankruptcy	Highest	Lowest
Risk	Low	High
Expected return	Low	High
Voting and control rights	No	Yes
Term	Dated	Perpetual
Share of total U.S. corporate financing (market values)	About 25%	About 75%

Bonds, Definitions

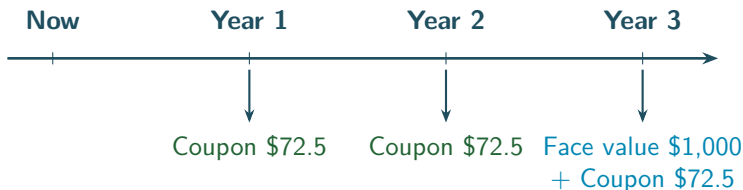
- **Bond**, a security that obligates the issuer (the borrower) to make specified payments to the investor (the lender, or “bondholder”)

Bonds, Definitions

- **Bond**, a security that obligates the issuer (the borrower) to make specified payments to the investor (the lender, or “bondholder”)
- Typically some periodic payments (the “coupon”), then a “face value” at the end

Bonds, Definitions

- **Bond**, a security that obligates the issuer (the borrower) to make specified payments to the investor (the lender, or “bondholder”)
- Typically some periodic payments (the “coupon”), then a “face value” at the end
- E.g., a three-year bond with an annual coupon of \$72.5 and a face value of \$1,000, the borrower promises to pay,



Bonds, Terminology

- **Face value (par value, principal)**, the final lump-sum payment at the “maturity” of the bond, usually \$1,000 per bond in the U.S.
 - **Not** the same as the value (price) of the bond

Bonds, Terminology

- **Face value (par value, principal)**, the final lump-sum payment at the “maturity” of the bond, usually \$1,000 per bond in the U.S.
 - **Not** the same as the value (price) of the bond
- **Maturity**, the time when the debt expires (“matures”) and the face value is paid
- **Time to maturity**, the number of years until maturity

Bonds, Terminology

- **Face value (par value, principal)**, the final lump-sum payment at the “maturity” of the bond, usually \$1,000 per bond in the U.S.
 - **Not** the same as the value (price) of the bond
- **Maturity**, the time when the debt expires (“matures”) and the face value is paid
- **Time to maturity**, the number of years until maturity
- **Coupon**, the periodic payments made to the bondholder before maturity

Bonds, Terminology

- **Face value (par value, principal)**, the final lump-sum payment at the “maturity” of the bond, usually \$1,000 per bond in the U.S.
 - **Not** the same as the value (price) of the bond
- **Maturity**, the time when the debt expires (“matures”) and the face value is paid
- **Time to maturity**, the number of years until maturity
- **Coupon**, the periodic payments made to the bondholder before maturity
- **Coupon rate**,

$$\text{Coupon rate} = \frac{\text{Annual coupon}}{\text{Face value}}$$

Bonds, Terminology

- Common fallacy, the coupon is not “interest”, and the coupon rate is not necessarily the interest rate

Bonds, Terminology

- Common fallacy, the coupon is not “interest”, and the coupon rate is not necessarily the interest rate
 - E.g., **Module 4**’s Puerto Rico zero-coupon bond, the coupon was 0 but the interest rate was 5.15%
 - Coupons can also be higher than the interest

Bonds, Terminology

- Common fallacy, the coupon is not “interest”, and the coupon rate is not necessarily the interest rate
 - E.g., **Module 4**’s Puerto Rico zero-coupon bond, the coupon was 0 but the interest rate was 5.15%
 - Coupons can also be higher than the interest
- **Zero-coupon bonds**, bonds that repay the principal at maturity with no coupon payments along the way

Bonds, Terminology

- Common fallacy, the coupon is not “interest”, and the coupon rate is not necessarily the interest rate
 - E.g., **Module 4**’s Puerto Rico zero-coupon bond, the coupon was 0 but the interest rate was 5.15%
 - Coupons can also be higher than the interest
- **Zero-coupon bonds**, bonds that repay the principal at maturity with no coupon payments along the way
- **Floating-rate bonds**, bonds whose coupon payments fluctuate with short-term interest rates (the coupon is not fixed)

Bonds vs. Loans, Main Differences

Bonds

- Sold to a broad group of investors, typically mutual funds, pension funds, insurance companies
- Subdivided into “smaller pieces”, usually \$1,000 face value
- Intended to be tradable
- Few “covenants” (restrictions) on the borrower

Loans

Bonds vs. Loans, Main Differences

Bonds

- Sold to a broad group of investors, typically mutual funds, pension funds, insurance companies
- Subdivided into “smaller pieces”, usually \$1,000 face value
- Intended to be tradable
- Few “covenants” (restrictions) on the borrower

Loans

- A single source of lending, typically a bank or a bank “syndicate” (group of banks)
- Not designed to be subdivided, although investors can still own “pieces” of a loan
- No active market, so harder to trade between investors
- Usually many covenants on the borrower

Other Types of Corporate Credit

- Beyond bonds and loans,
 - Revolving credit facilities (like a corporate credit card)
 - Commercial paper (short-term bonds, also called “notes”)

Other Types of Corporate Credit

- Beyond bonds and loans,
 - Revolving credit facilities (like a corporate credit card)
 - Commercial paper (short-term bonds, also called “notes”)
- The way to value all kinds of corporate debt is the same, **Module 4**’s time-value-of-money principles

Bonds, Pricing

- **Bond price (value)**, the present value of all its cash flows (coupons and face value), discounted at the “required rate of return” r

Bonds, Pricing

- **Bond price (value)**, the present value of all its cash flows (coupons and face value), discounted at the “required rate of return” r
- E.g., for a risk-free bond with time to maturity T ,

$$\text{Price} = \text{PV} = \frac{\text{Coupon}_1}{(1+r)^1} + \frac{\text{Coupon}_2}{(1+r)^2} + \dots + \frac{\text{Coupon}_T + \text{Face value}}{(1+r)^T}$$

Bonds, Pricing a Risky Bond

- Risky bond, discount the **expected** coupon and face value payments, not the promised ones
 - The expectation accounts for the probability that the borrower does not pay

$$\text{Price} = \text{PV} = \frac{E[\text{Coupon}_1]}{(1+r)^1} + \frac{E[\text{Coupon}_2]}{(1+r)^2} + \dots + \frac{E[\text{Coupon}_T + \text{Face value}]}{(1+r)^T}$$

Bonds, Pricing a Risky Bond

- Risky bond, discount the **expected** coupon and face value payments, not the promised ones

- The expectation accounts for the probability that the borrower does not pay

$$\text{Price} = \text{PV} = \frac{E[\text{Coupon}_1]}{(1+r)^1} + \frac{E[\text{Coupon}_2]}{(1+r)^2} + \dots + \frac{E[\text{Coupon}_T + \text{Face value}]}{(1+r)^T}$$

- The required rate of return r is set by the opportunity cost, “what could I get investing in something else with a similar risk”
 - Depends on the time value of money and the riskiness of the bond (more on this a bit later)

How to Calculate Expectations?

- How to calculate $E[\text{Coupon}_1]$? What is it exactly?

How to Calculate Expectations?

- How to calculate $E[\text{Coupon}_1]$? What is it exactly?
- The coupon is only *promised*, if the borrower defaults, it may pay only part of the coupon, or nothing

How to Calculate Expectations?

- How to calculate $E[\text{Coupon}_1]$? What is it exactly?
- The coupon is only *promised*, if the borrower defaults, it may pay only part of the coupon, or nothing
- **Expectation operator** $E[\cdot]$, the probability-weighted average payment the borrower will pay (which is also what the investor will receive)

Example, Expected Coupon

- Suppose in the first year, the borrower will pay,
 - \$72.50 (the full coupon) with 60% probability
 - \$50 with 20% probability (default but high “recovery”)
 - \$0 with 20% probability (default and no “recovery”)
- What is $E[\text{Coupon}_1]$?

Example, Expected Coupon

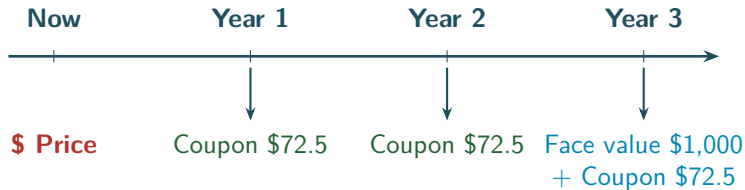
- Suppose in the first year, the borrower will pay,
 - \$72.50 (the full coupon) with 60% probability
 - \$50 with 20% probability (default but high “recovery”)
 - \$0 with 20% probability (default and no “recovery”)
- What is $E[\text{Coupon}_1]$?
- Solution, the expected coupon is the probability-weighted average amount,
$$E[\text{Coupon}_1] = 0.6 \times 72.50 + 0.2 \times 50 + 0.2 \times 0 = \$53.50$$

Example, Expected Coupon

- Suppose in the first year, the borrower will pay,
 - \$72.50 (the full coupon) with 60% probability
 - \$50 with 20% probability (default but high “recovery”)
 - \$0 with 20% probability (default and no “recovery”)
- What is $E[\text{Coupon}_1]$?
- Solution, the expected coupon is the probability-weighted average amount,
$$E[\text{Coupon}_1] = 0.6 \times 72.50 + 0.2 \times 50 + 0.2 \times 0 = \$53.50$$
- This is what goes in the numerator when discounting (risky) bonds

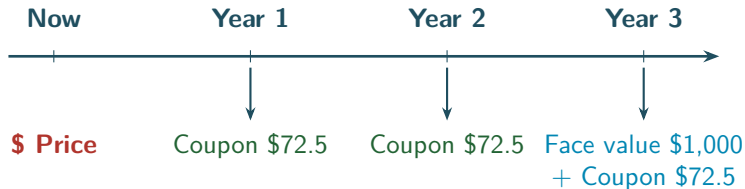
Example, Bond Pricing

- A (risk-free) bond has an annual coupon of \$72.5 and a face value of \$1,000, with three years to maturity



Example, Bond Pricing

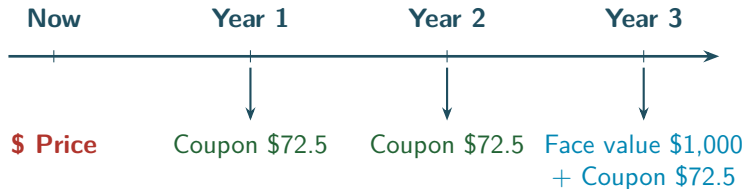
- A (risk-free) bond has an annual coupon of \$72.5 and a face value of \$1,000, with three years to maturity



- What is the price of the bond if the required rate of return r is 4%?

Example, Bond Pricing

- A (risk-free) bond has an annual coupon of \$72.5 and a face value of \$1,000, with three years to maturity



- What is the price of the bond if the required rate of return r is 4%?

$$\text{Price} = \frac{\$72.50}{(1 + 0.04)^1} + \frac{\$72.50}{(1 + 0.04)^2} + \frac{\$72.50 + \$1,000}{(1 + 0.04)^3} = \$1,090.19$$

Example, Bond Pricing with a Financial Calculator

- The same bond, an annual coupon of \$72.5, a face value of \$1,000, three years to maturity, and a required rate of return of 4%
- Use a financial calculator to price the bond

Example, Bond Pricing with a Financial Calculator

- The same bond, an annual coupon of \$72.5, a face value of \$1,000, three years to maturity, and a required rate of return of 4%
- Use a financial calculator to price the bond
- Solution,
 - 1000 **FV**
 - 72.5 **PMT**
 - 3 **N**
 - 4 **I/Y**
 - **CPT**, then **PV**

Example, Bond Pricing with a Financial Calculator

- The same bond, an annual coupon of \$72.5, a face value of \$1,000, three years to maturity, and a required rate of return of 4%
- Use a financial calculator to price the bond
- Solution,
 - 1000 **FV**
 - 72.5 **PMT**
 - 3 **N**
 - 4 **I/Y**
 - **CPT**, then **PV**
- The output is $-\$1,090.19$ (why the minus sign? recall how financial calculators get to the answer in [Module 4](#))

Valuing Bonds with a Financial Calculator

- The financial calculator keys map onto a bond,
 - **N**, the number of periods (years) until maturity
 - **PV**, the price of the bond (the present value of the coupons and the face value)
 - **PMT**, the coupons (in dollars)
 - **FV**, the face value (the lump sum paid at the end)
 - **I/Y**, the discount rate, when **PMT** and **FV** are expected cash flows it is the required rate of return

Bond Prices Are Often Quoted as a Percentage of Par

- Bond prices are often quoted as a percentage of “par” (recall, par value is another name for the face value)

Bond Prices Are Often Quoted as a Percentage of Par

- Bond prices are often quoted as a percentage of “par” (recall, par value is another name for the face value)
- E.g., the bond we just analyzed, face value \$1,000 and price \$1,090.19, could be quoted as,

$$\text{Price} = 109.019\%$$

Recap, Bond Pricing

- Companies issue bonds (and other debt) to raise financing from investors

Recap, Bond Pricing

- Companies issue bonds (and other debt) to raise financing from investors
- In return, the companies make pre-specified payments of coupons and face value to the investors

Recap, Bond Pricing

- Companies issue bonds (and other debt) to raise financing from investors
- In return, the companies make pre-specified payments of coupons and face value to the investors
- The price (value) of a bond is the discounted (expected) payments, at the required rate of return

Bond Yields

- **Yield to maturity (YTM)**, the discount rate y for which the present value of the bond's promised payments equals its price (also simply the “yield”)

Bond Yields

- **Yield to maturity (YTM)**, the discount rate y for which the present value of the bond's promised payments equals its price (also simply the “yield”)
- Given the price of a bond (PV), its maturity T , and its coupons, find the yield to maturity by solving for y ,

$$\text{Price} = \frac{\text{Coupon}}{(1+y)^1} + \frac{\text{Coupon}}{(1+y)^2} + \cdots + \frac{\text{Coupon} + \text{Face value}}{(1+y)^T}$$

Example, Bond Yields

- Solving for the YTM analytically is hard, so use the financial calculator
- A bond's annual coupon rate is 7.25%. Its face value is \$1,000. Its maturity is 3 years. Its current market price is \$1,090.19. What is its YTM?

Example, Bond Yields

- Solving for the YTM analytically is hard, so use the financial calculator
- A bond's annual coupon rate is 7.25%. Its face value is \$1,000. Its maturity is 3 years. Its current market price is \$1,090.19. What is its YTM?
- Solution,
 - 1000 **FV**
 - 72.5 **PMT**
 - 3 **N**
 - -1090.19 **PV**
 - **CPT**, then **I/Y**

Example, Bond Yields

- Solving for the YTM analytically is hard, so use the financial calculator
- A bond's annual coupon rate is 7.25%. Its face value is \$1,000. Its maturity is 3 years. Its current market price is \$1,090.19. What is its YTM?
- Solution,
 - 1000 **FV**
 - 72.5 **PMT**
 - 3 **N**
 - -1090.19 **PV**
 - **CPT**, then **I/Y**
- The yield is 4%

Bond Prices Are Often Quoted as Yields

- For each bond, price and yield have a direct one-for-one relationship

U.S. 10 Year Treasury

US10Y:Tradeweb

RT Quote | Exchange

Yield | 5:05 PM EST

1.822% ▼ -0.162

1D 5D 1M 3M 6M YTD 1Y 5Y ALL



SUMMARY

NEWS

KEY STATS

Yield Open	1.892%	Price	100.4844	Price Day High	100.5312
Yield Day High	1.927%	Price Change	+1.4688	Price Day Low	99.5312
Yield Day Low	1.816%	Price Change %	+1.4844%	Coupon	1.875%
Yield Prev Close	1.984%	Price Prev Close	99.0156	Maturity	2032-02-15

LATEST ON U.S. 10 YEAR TREASURY

- Oil prices could determine how markets react to Russia's Ukraine invasion 49 MIN AGO - CNBC.COM
- Treasury yields slide as fresh sanctions are imposed on Russia 13 HOURS AGO - CNBC.COM

Bond Prices Are Often Quoted as Yields

- For each bond, price and yield have a direct one-for-one relationship
- So bond prices are often quoted not in dollars or percent of par, but as the “yield”

U.S. 10 Year Treasury

US10Y:Tradeweb

RT Quote | Exchange

Yield | 5:05 PM EST

1.822% ▼ **-0.162**

1D 5D 1M 3M 6M YTD 1Y 5Y ALL

+ Comparison

1D

Display

Studies

Settings

Tools

Chart



SUMMARY

NEWS

KEY STATS

Yield Open	1.892%	Price	100.4844	Price Day High	100.5312
Yield Day High	1.927%	Price Change	+1.4688	Price Day Low	99.5312
Yield Day Low	1.816%	Price Change %	+1.4844%	Coupon	1.875%
Yield Prev Close	1.984%	Price Prev Close	99.0156	Maturity	2032-02-15

LATEST ON U.S. 10 YEAR TREASURY

- Oil prices could determine how markets react to Russia's Ukraine invasion 49 MIN AGO - CNBC.COM
- Treasury yields slide as fresh sanctions are imposed on Russia 13 HOURS AGO - CNBC.COM

Bond Prices Are Often Quoted as Yields

- For each bond, price and yield have a direct one-for-one relationship
- So bond prices are often quoted not in dollars or percent of par, but as the “yield”
- E.g., the 10-year U.S. Treasury bond is quoted as a yield of 1.822% (its price is 100.48%)

U.S. 10 Year Treasury

US10Y:Tradeweb

RT Quote | Exchange

Yield | 5:05 PM EST

1.822% ▼ -0.162

1D 5D 1M 3M 6M YTD 1Y 5Y ALL

+ Comparison

1D

Display

Studies

Settings

Tools

Share

Fullscreen



SUMMARY

NEWS

KEY STATS

Yield Open	1.892%	Price	100.4844	Price Day High	100.5312
Yield Day High	1.927%	Price Change	+1.4688	Price Day Low	99.5312
Yield Day Low	1.816%	Price Change %	+1.4844%	Coupon	1.875%
Yield Prev Close	1.984%	Price Prev Close	99.0156	Maturity	2032-02-15

LATEST ON U.S. 10 YEAR TREASURY

- Oil prices could determine how markets react to Russia's Ukraine invasion 49 MIN AGO - CNBC.COM
- Treasury yields slide as fresh sanctions are imposed on Russia 13 HOURS AGO - CNBC.COM

Is the YTM the Same as the Required Rate of Return r ?

- Yield, the return you would earn if you buy a bond, hold it until maturity, and it does not default

Is the YTM the Same as the Required Rate of Return r ?

- Yield, the return you would earn if you buy a bond, hold it until maturity, and it does not default
- For risk-free bonds, YTM and r are the same

Is the YTM the Same as the Required Rate of Return r ?

- Yield, the return you would earn if you buy a bond, hold it until maturity, and it does not default
- For risk-free bonds, YTM and r are the same
- For risky bonds, YTM and r are different (specifically, $\text{YTM} > r$)

YTM vs. r , Promised vs. Expected Payments

- When calculating r , use **expected** payments, $E[\cdot]$ accounts for the probability that the borrower pays less or nothing,

$$\text{Price} = \frac{E[\text{Coupon}]}{(1+r)^1} + \frac{E[\text{Coupon}]}{(1+r)^2} + \dots + \frac{E[\text{Coupon} + \text{Face value}]}{(1+r)^T}$$

YTM vs. r , Promised vs. Expected Payments

- When calculating r , use **expected** payments, $E[\cdot]$ accounts for the probability that the borrower pays less or nothing,

$$\text{Price} = \frac{E[\text{Coupon}]}{(1+r)^1} + \frac{E[\text{Coupon}]}{(1+r)^2} + \dots + \frac{E[\text{Coupon} + \text{Face value}]}{(1+r)^T}$$

- When calculating the YTM, always use **promised** payments, even for risky bonds,

$$\text{Price} = \frac{\text{Coupon}}{(1+y)^1} + \frac{\text{Coupon}}{(1+y)^2} + \dots + \frac{\text{Coupon} + \text{Face value}}{(1+y)^T}$$

Example, YTM vs. r on a Financial Calculator

- A one-year zero-coupon bond has a probability of default (the issuer does not pay the face value) of $p_{\text{default}} = 10\%$. Its face value is \$1,000. Its price is \$850
- What are the yield y and the expected (required) return r ?

Example, YTM vs. r on a Financial Calculator

- A one-year zero-coupon bond has a probability of default (the issuer does not pay the face value) of $p_{\text{default}} = 10\%$. Its face value is \$1,000. Its price is \$850
- What are the yield y and the expected (required) return r ?
- A. The yield, using the **promised** cash flow

N	I/Y	PV	PMT	FV
1	y	-850	0	1,000

Example, YTM vs. r on a Financial Calculator

- A one-year zero-coupon bond has a probability of default (the issuer does not pay the face value) of $p_{\text{default}} = 10\%$. Its face value is \$1,000. Its price is \$850
- What are the yield y and the expected (required) return r ?
- A. The yield, using the **promised** cash flow

N	I/Y	PV	PMT	FV
1	y	-850	0	1,000

- YTM is 17.65%

Example, YTM vs. r on a Financial Calculator

- A one-year zero-coupon bond has a probability of default (the issuer does not pay the face value) of $p_{\text{default}} = 10\%$. Its face value is \$1,000. Its price is \$850
- What are the yield y and the expected (required) return r ?

- A. The yield, using the **promised** cash flow

N	I/Y	PV	PMT	FV
1	y	-850	0	1,000

- YTM is 17.65%

- B. The required return, using the **expected** cash flow,
 $(1 - p_{\text{default}}) \times \text{FV} = 900$

N	I/Y	PV	PMT	FV
1	r	-850	0	900

Example, YTM vs. r on a Financial Calculator

- A one-year zero-coupon bond has a probability of default (the issuer does not pay the face value) of $p_{\text{default}} = 10\%$. Its face value is \$1,000. Its price is \$850
- What are the yield y and the expected (required) return r ?

- A. The yield, using the **promised** cash flow

N	I/Y	PV	PMT	FV
1	y	-850	0	1,000

- YTM is 17.65%

- B. The required return, using the **expected** cash flow,
 $(1 - p_{\text{default}}) \times \text{FV} = 900$

N	I/Y	PV	PMT	FV
1	r	-850	0	900

- r is 5.88%

Bond Premiums and Discounts

- A bond priced **above** its face value trades at a **premium**
 - Happens if coupon rate $>$ yield to maturity

Bond Premiums and Discounts

- A bond priced **above** its face value trades at a **premium**
 - Happens if coupon rate $>$ yield to maturity
- A bond priced **below** its face value trades at a **discount**
 - Happens if coupon rate $<$ yield to maturity

Bond Premiums and Discounts

- A bond priced **above** its face value trades at a **premium**
 - Happens if coupon rate $>$ yield to maturity
- A bond priced **below** its face value trades at a **discount**
 - Happens if coupon rate $<$ yield to maturity
- A bond priced **equal** to its face value trades at **par**
 - When will this happen?

Bond Premiums and Discounts

- A bond priced **above** its face value trades at a **premium**
 - Happens if coupon rate $>$ yield to maturity
- A bond priced **below** its face value trades at a **discount**
 - Happens if coupon rate $<$ yield to maturity
- A bond priced **equal** to its face value trades at **par**
 - When will this happen?
 - Happens if coupon rate $=$ yield to maturity

Example, Bond Premiums and Discounts

- A (risk-free) bond pays annual coupons. Its face value is \$1,000. Its time to maturity is five years. Its yield is 5%
- What is the price of the bond if the annual coupon rate is,
 1. 7%?
 2. 3%?
 3. 5%?

Example, Bond Premiums and Discounts

- A (risk-free) bond pays annual coupons. Its face value is \$1,000. Its time to maturity is five years. Its yield is 5%
- What is the price of the bond if the annual coupon rate is,
 1. 7%?
 2. 3%?
 3. 5%?
- Solution, 1. if the coupon rate is 7%, the coupons are $7\% \times \$1,000 = \70 per year,
 - 1000 **FV**, 5 **N**, 5 **I/Y**, 70 **PMT**, then **CPT** **PV**

Example, Bond Premiums and Discounts

- A (risk-free) bond pays annual coupons. Its face value is \$1,000. Its time to maturity is five years. Its yield is 5%
- What is the price of the bond if the annual coupon rate is,
 1. 7%?
 2. 3%?
 3. 5%?
- Solution, 1. if the coupon rate is 7%, the coupons are $7\% \times \$1,000 = \70 per year,
 - 1000 **FV**, 5 **N**, 5 **I/Y**, 70 **PMT**, then **CPT** **PV**
- The price is \$1,087, a “premium bond”

Example, Bond Premiums and Discounts

- A (risk-free) bond pays annual coupons. Its face value is \$1,000. Its time to maturity is five years. Its yield is 5%
- What is the price of the bond if the annual coupon rate is,
 1. 7%?
 2. 3%?
 3. 5%?
- Solution, 1. if the coupon rate is 7%, the coupons are $7\% \times \$1,000 = \70 per year,
 - 1000 **FV**, 5 **N**, 5 **I/Y**, 70 **PMT**, then **CPT** **PV**
- The price is \$1,087, a “premium bond”
- Bonds 2 and 3 are worth \$913 (a “discount bond”) and \$1,000 (“par”), respectively

Practice Problem

- The table shows data for three bonds. Each bond's coupon is zero. Each bond's face value is \$1,000

Bond	Price	Maturity (years)	Yield to maturity
A	\$310	25	?
B	\$310	?	9%
C	?	11	8%

- What is the yield to maturity of bond A?
- What is the maturity of bond B?
- What is the price of bond C?

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
a. Bond A	25	?	-310	0	1,000
b. Bond B	?	9	-310	0	1,000
c. Bond C	11	8	?	0	-1,000

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
a. Bond A	25	4.796	-310	0	1,000
b. Bond B	?	9	-310	0	1,000
c. Bond C	11	8	?	0	-1,000

- a. The yield to maturity of bond A is 4.796%

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
a. Bond A	25	4.796	-310	0	1,000
b. Bond B	13.59	9	-310	0	1,000
c. Bond C	11	8	?	0	-1,000

- a. The yield to maturity of bond A is 4.796%
- b. The maturity of bond B is 13.59 years

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
a. Bond A	25	4.796	-310	0	1,000
b. Bond B	13.59	9	-310	0	1,000
c. Bond C	11	8	428.88	0	-1,000

- a. The yield to maturity of bond A is 4.796%
- b. The maturity of bond B is 13.59 years
- c. The price of bond C is \$428.88

Practice Problem

- A firm's bonds pay an annual coupon rate of 6.8%. Their face value is \$1,000. Their price is \$795.69. Their yield to maturity is 9.50%
- What is the remaining maturity of these bonds?

Practice Problem

- Solution, the bond price is the PV of the coupon payments plus the PV of the face value,

Bond price = PV of coupon payments + PV of face value

Bond price = $C \times ((1 \div r) - \{1 \div [r(1 + r)^t]\}) + FV \div (1 + r)^t$

$\$795.69 = (0.068 \times \$1,000) \times ((1 \div 0.095) - \{1 \div [0.095 \times (1 + 0.095)^t]\}) + \$1,000 \div (1.095)^t$

Practice Problem

- Solution, the bond price is the PV of the coupon payments plus the PV of the face value,

Bond price = PV of coupon payments + PV of face value

Bond price = $C \times ((1 \div r) - \{1 \div [r(1 + r)^t]\}) + FV \div (1 + r)^t$

$$\$795.69 = (0.068 \times \$1,000) \times ((1 \div 0.095) - \{1 \div [0.095 \times (1 + 0.095)^t]\}) + \$1,000 \div (1.095)^t$$

- On the financial calculator,
 - 9.5 **I/Y**, -795.69 **PV**, 68 **PMT**, 1000 **FV**, then **CPT** **N**

Practice Problem

- Solution, the bond price is the PV of the coupon payments plus the PV of the face value,

Bond price = PV of coupon payments + PV of face value

Bond price = $C \times ((1 \div r) - \{1 \div [r(1 + r)^t]\}) + FV \div (1 + r)^t$

$$\$795.69 = (0.068 \times \$1,000) \times ((1 \div 0.095) - \{1 \div [0.095 \times (1 + 0.095)^t]\}) + \$1,000 \div (1.095)^t$$

- On the financial calculator,
 - 9.5 **I/Y**, -795.69 **PV**, 68 **PMT**, 1000 **FV**, then **CPT** **N**
- The remaining maturity is about 14 years ($N = 13.98$)

What Determines r ?

- What determines r (the required rate of return, or expected return)?

What Determines r ?

- What determines r (the required rate of return, or expected return)?
- 1. Prevailing (risk-free) interest rates (the opportunity cost for time value)
 - Higher prevailing (risk-free) interest rate, higher r

What Determines r ?

- What determines r (the required rate of return, or expected return)?
 1. Prevailing (risk-free) interest rates (the opportunity cost for time value)
 - Higher prevailing (risk-free) interest rate, higher r
 2. Risk of default (the opportunity cost for risk-taking)
 - Higher (systematic) risk of default, higher r

Determinants of the Prevailing Interest Rates

- The interest rate is a price, it sets the demand for borrowing equal to the supply of lending

Determinants of the Prevailing Interest Rates

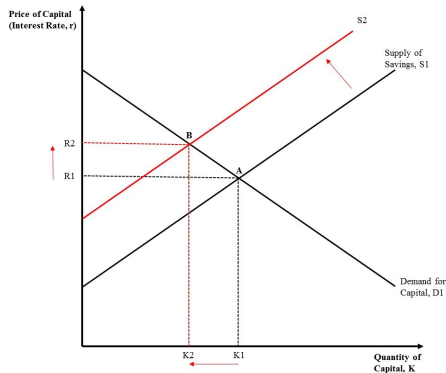
- The interest rate is a price, it sets the demand for borrowing equal to the supply of lending
- Not enough supply, or too much demand, for “renting money”, interest rates rise (higher r on all bonds), or vice versa

Determinants of the Prevailing Interest Rates

- The interest rate is a price, it sets the demand for borrowing equal to the supply of lending
- Not enough supply, or too much demand, for “renting money”, interest rates rise (higher r on all bonds), or vice versa
- The Federal Reserve does not “set” the interest rate, it tilts supply and demand by buying and selling bonds (creating “money” to buy them)

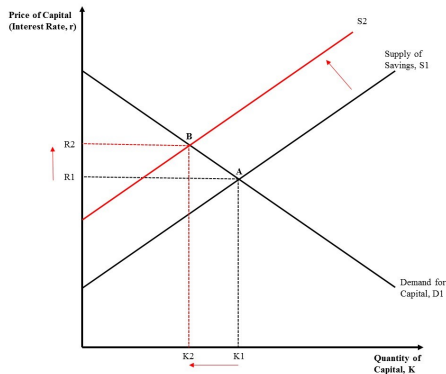
Example, Negative Supply Shock

- The market starts at equilibrium point A, interest rate R_1 and quantity of lending K_1



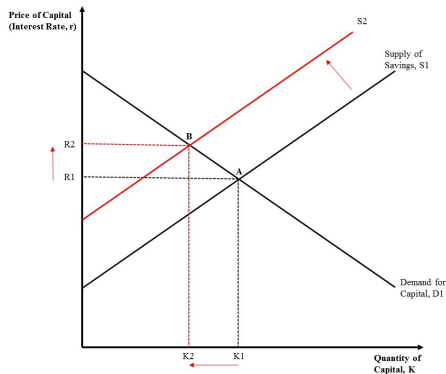
Example, Negative Supply Shock

- The market starts at equilibrium point A, interest rate R_1 and quantity of lending K_1
- A negative supply shock, fewer people want to lend (or the Federal Reserve cuts back on creating money)
- The price (the interest rate) rises to R_2 , lending and borrowing contract to K_2 (point B)



Example, Negative Supply Shock

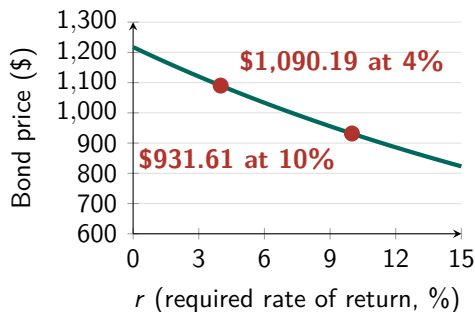
- The market starts at equilibrium point A, interest rate R_1 and quantity of lending K_1
- A negative supply shock, fewer people want to lend (or the Federal Reserve cuts back on creating money)
- The price (the interest rate) rises to R_2 , lending and borrowing contract to K_2 (point B)
- See also [Investopedia, Forces Behind Interest Rates](#)



Interest Rates and Bond Prices Move in Opposite Directions

- Because the bond price is,

$$\text{Price} = \frac{E[\text{Coupon}]}{(1+r)^1} + \dots + \frac{E[\text{Coupon} + \text{Face value}]}{(1+r)^T}$$

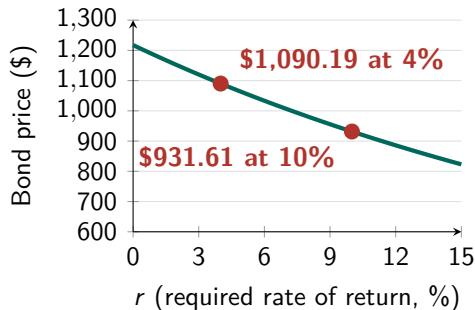


Interest Rates and Bond Prices Move in Opposite Directions

- Because the bond price is,

$$\text{Price} = \frac{E[\text{Coupon}]}{(1+r)^1} + \dots + \frac{E[\text{Coupon} + \text{Face value}]}{(1+r)^T}$$

- Higher r , lower bond price (the chart shows the running example bond)

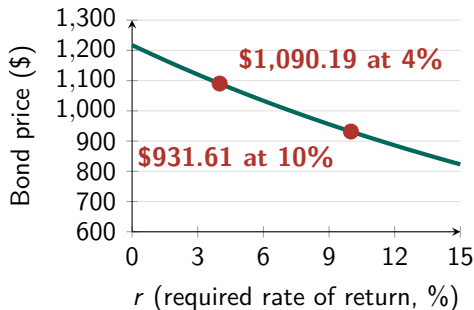


Interest Rates and Bond Prices Move in Opposite Directions

- Because the bond price is,

$$\text{Price} = \frac{E[\text{Coupon}]}{(1+r)^1} + \dots + \frac{E[\text{Coupon} + \text{Face value}]}{(1+r)^T}$$

- Higher r , lower bond price (the chart shows the running example bond)
- So higher interest rates, higher r , lower bond prices

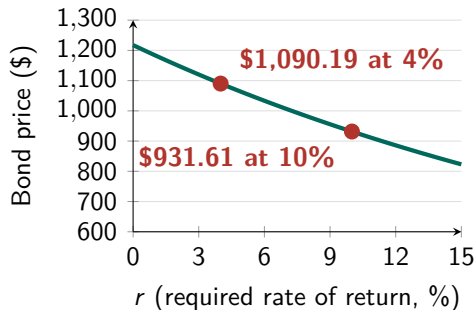


Interest Rates and Bond Prices Move in Opposite Directions

- Because the bond price is,

$$\text{Price} = \frac{E[\text{Coupon}]}{(1+r)^1} + \dots + \frac{E[\text{Coupon} + \text{Face value}]}{(1+r)^T}$$

- Higher r , lower bond price (the chart shows the running example bond)
- So higher interest rates, higher r , lower bond prices
- Interest rates and bond prices move in opposite directions!



Default Risk

- **Default risk (credit risk)**, the risk that a bond issuer may default on its bonds

Default Risk

- **Default risk (credit risk)**, the risk that a bond issuer may default on its bonds
- **Default premium**, the additional yield on a bond that investors require for bearing credit risk

Default Risk, Credit Ratings

- Credit rating agencies provide ratings intended to capture default risk
 - Largest, Moody's, Standard & Poor's, and Fitch

Default Risk, Credit Ratings

- Credit rating agencies provide ratings intended to capture default risk
 - Largest, Moody's, Standard & Poor's, and Fitch
- Bond ratings depend on,
 1. The probability of default
 2. The expected loss in default

Default Risk, Credit Ratings

- Credit rating agencies provide ratings intended to capture default risk
 - Largest, Moody's, Standard & Poor's, and Fitch
- Bond ratings depend on,
 1. The probability of default
 2. The expected loss in default
- **Investment-grade bonds**, rated Baa or higher by Moody's, or BBB or higher by Standard & Poor's

Default Risk, Credit Ratings

- Credit rating agencies provide ratings intended to capture default risk
 - Largest, Moody's, Standard & Poor's, and Fitch
- Bond ratings depend on,
 1. The probability of default
 2. The expected loss in default
- **Investment-grade bonds**, rated Baa or higher by Moody's, or BBB or higher by Standard & Poor's
- **High-yield bonds** ("junk" bonds), rated Ba or BB and lower

Default Risk, Credit Ratings

Moody's	S&P	Default within 10 yr	Safety
Investment-grade bonds			
Aaa	AAA	0.1%	The strongest rating, ability to repay interest and principal is very strong
Aa	AA	0.7%	Very strong likelihood that interest and principal will be repaid
A	A	2.2%	Strong ability to repay, but some vulnerability to changes in circumstances
Baa	BBB	3.5%	Adequate capacity to repay, more vulnerability to changes in economic circumstances

Source, Brealey, Myers, and Marcus, Table 6.2

Default Risk, Credit Ratings

Moody's	S&P	Default within 10 yr	Safety
High-yield bonds			
Ba	BB	15.7%	Considerable uncertainty about ability to repay
B	B	35.5%	Likelihood of interest and principal payments over sustained periods is questionable
Caa, Ca	CCC, CC	48.9%	Bonds that may already be in default or in danger of imminent default
C	C	n/a	Little prospect for interest or principal on the debt ever to be repaid

Source, Brealey, Myers, and Marcus, Table 6.2

Default Risk and Bond Yields

Issuer	Ticker	Coupon (%)	Maturity	Rating	Yield (%)
Microsoft	MSFT	2.40	2022	AAA	2.89
Walmart	WMT	6.75	2023	AA	2.92
Bristol Myers	BMJ	6.80	2023	A	3.43
Corning	GLW	7.73	2024	BBB	4.00
Southwestern Energy	SWN	4.10	2022	BB	4.80
AMD	AMD	7.00	2024	B	5.28

Source, FINRA, senior corporate bonds, March 2018 (Brealey, Myers, and Marcus, Table 6.3)

Default Risk and Bond Yields

Issuer	Ticker	Coupon (%)	Maturity	Rating	Yield (%)
Microsoft	MSFT	2.40	2022	AAA	2.89
Walmart	WMT	6.75	2023	AA	2.92
Bristol Myers	BMJ	6.80	2023	A	3.43
Corning	GLW	7.73	2024	BBB	4.00
Southwestern Energy	SWN	4.10	2022	BB	4.80
AMD	AMD	7.00	2024	B	5.28

Source, FINRA, senior corporate bonds, March 2018 (Brealey, Myers, and Marcus, Table 6.3)

- Yields tend to be higher (prices lower) for lower-rated bonds

Rate of Return of Holding a Bond

- **Rate of return** (of holding a bond), the investor's realized return over a certain period

$$\text{Rate of return} = \frac{\text{Coupon income} + \text{Price change}}{\text{Investment}}$$

Rate of Return of Holding a Bond

- **Rate of return** (of holding a bond), the investor's realized return over a certain period

$$\text{Rate of return} = \frac{\text{Coupon income} + \text{Price change}}{\text{Investment}}$$

- This is what you earn “ex post”, different from the yield and from the required (expected) return

Rate of Return of Holding a Bond

- **Rate of return** (of holding a bond), the investor's realized return over a certain period

$$\text{Rate of return} = \frac{\text{Coupon income} + \text{Price change}}{\text{Investment}}$$

- This is what you earn “ex post”, different from the yield and from the required (expected) return
- It is the “holding period return” of the bond, as in [Module 4](#)

Example, Rate of Return of Holding a Bond

- You purchase an 8% annual coupon, 20-year maturity bond when its YTM is 9%
- You sell it one year later, when the YTM is 10%. What is your rate of return over the year?

Example, Rate of Return of Holding a Bond

- You purchase an 8% annual coupon, 20-year maturity bond when its YTM is 9%
- You sell it one year later, when the YTM is 10%. What is your rate of return over the year?
- Solution, first use the financial calculator to find the prices at $t = 0$ and $t = 1$,
 - $P_0 = 908.71$ (20 **N**, 80 **PMT**, 1000 **FV**, 9 **I/Y**, then **CPT** **PV**)
 - $P_1 = 832.70$ (19 **N**, 80 **PMT**, 1000 **FV**, 10 **I/Y**, then **CPT** **PV**)

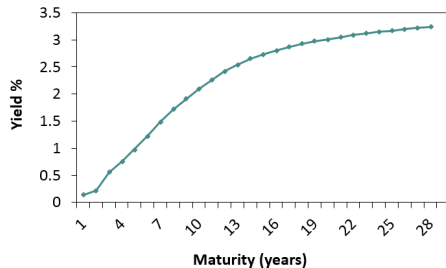
Example, Rate of Return of Holding a Bond

- You purchase an 8% annual coupon, 20-year maturity bond when its YTM is 9%
- You sell it one year later, when the YTM is 10%. What is your rate of return over the year?
- Solution, first use the financial calculator to find the prices at $t = 0$ and $t = 1$,
 - $P_0 = 908.71$ (20 **N**, 80 **PMT**, 1000 **FV**, 9 **I/Y**, then **CPT** **PV**)
 - $P_1 = 832.70$ (19 **N**, 80 **PMT**, 1000 **FV**, 10 **I/Y**, then **CPT** **PV**)
- Then,

$$\text{Rate of return} = \frac{\text{Coupon income} + \text{Price change}}{\text{Investment}} = \frac{80 + (832.70 - 908.71)}{908.71} = 0.44\%$$

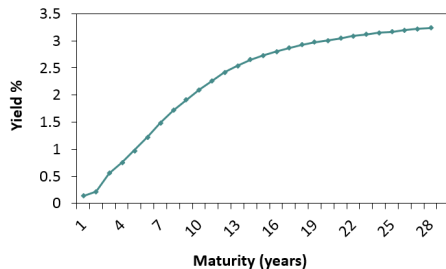
The Yield Curve

- **Term structure of interest rates**, how interest rates vary across bond maturities



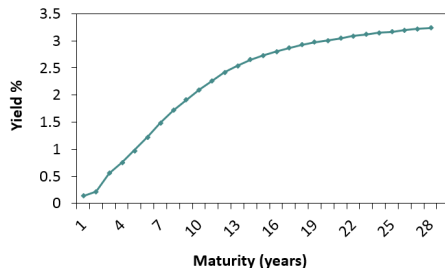
The Yield Curve

- **Term structure of interest rates**, how interest rates vary across bond maturities
- **Yield curve**, a plot of yields to maturity (YTM) against time to maturity



The Yield Curve

- **Term structure of interest rates**, how interest rates vary across bond maturities
- **Yield curve**, a plot of yields to maturity (YTM) against time to maturity
- The yield curve is usually (but not always) upward-sloping



Recap of Part 1

- Bond price, the present value of its coupons and face value, discounted at the required rate of return r

Recap of Part 1

- Bond price, the present value of its coupons and face value, discounted at the required rate of return r
- The yield to maturity y discounts the **promised** payments, r discounts the **expected** payments

Recap of Part 1

- Bond price, the present value of its coupons and face value, discounted at the required rate of return r
- The yield to maturity y discounts the **promised** payments, r discounts the **expected** payments
- Coupon rate above, below, or equal to the yield means a premium, discount, or par bond

Recap of Part 1

- Bond price, the present value of its coupons and face value, discounted at the required rate of return r
- The yield to maturity y discounts the **promised** payments, r discounts the **expected** payments
- Coupon rate above, below, or equal to the yield means a premium, discount, or par bond
- Interest rates and bond prices move in opposite directions

Recap of Part 1

- Bond price, the present value of its coupons and face value, discounted at the required rate of return r
- The yield to maturity y discounts the **promised** payments, r discounts the **expected** payments
- Coupon rate above, below, or equal to the yield means a premium, discount, or par bond
- Interest rates and bond prices move in opposite directions
- r rises with prevailing interest rates and with default risk, lower-rated bonds carry higher yields

Module 5, Part 2

Stocks

Goals

- **Part 1, bonds**
 - Differences between debt and equity
 - Valuation
 - Yields
 - Returns and interest rates
 - Default risk
 - Special features of bond securities
- **Part 2, stocks**
 - Valuation, dividend discount model
 - Valuation, comparables
 - Market efficiency

Valuing Stocks

- Two alternative ways to value stocks,
 1. Dividend discount model
 2. Valuation by comparables

Valuing Stocks

- Two alternative ways to value stocks,
 1. Dividend discount model
 2. Valuation by comparables
- The approaches are complementary, each has advantages and disadvantages, and no approach provides “the” right answer
- Professionals use a combination

Valuing Stocks

- Two alternative ways to value stocks,
 1. Dividend discount model
 2. Valuation by comparables
- The approaches are complementary, each has advantages and disadvantages, and no approach provides “the” right answer
- Professionals use a combination
- You will learn more in “FINE 4100 Advanced Financial Management” and “FINE 4110 Investments in Equities”

Stock Valuation, Terminology

- **Dividend**, a periodic cash distribution from the firm to its shareholders

Stock Valuation, Terminology

- **Dividend**, a periodic cash distribution from the firm to its shareholders
- **Dividend yield**, $\frac{\text{Dividend}}{\text{Stock price}}$

Stock Valuation, Terminology

- **Dividend**, a periodic cash distribution from the firm to its shareholders
- **Dividend yield**, $\frac{\text{Dividend}}{\text{Stock price}}$
- **Earnings per share (EPS)**, $\frac{\text{Net income}}{\text{Number of shares outstanding}}$

Stock Valuation, Terminology

- **Dividend**, a periodic cash distribution from the firm to its shareholders
- **Dividend yield**, $\frac{\text{Dividend}}{\text{Stock price}}$
- **Earnings per share (EPS)**, $\frac{\text{Net income}}{\text{Number of shares outstanding}}$
- **P/E ratio**, $\frac{\text{Stock price}}{\text{Earnings per share}}$

Valuing Stocks, Dividend Discount Model

- Suppose you plan to buy a stock and own it forever
- What price should you pay for the stock today (P_0)?

Valuing Stocks, Dividend Discount Model

- Suppose you plan to buy a stock and own it forever
- What price should you pay for the stock today (P_0)?
- Stock value, the present value of the cash flows you get from owning it,

$$P_0 = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \frac{\text{Div}_3}{(1 + r_e)^3} + \cdots + \frac{\text{Div}_\infty}{(1 + r_e)^\infty} = \text{PV}(\text{Dividends})$$

Valuing Stocks, Dividend Discount Model

- Suppose you plan to buy a stock and own it forever
- What price should you pay for the stock today (P_0)?
- Stock value, the present value of the cash flows you get from owning it,

$$P_0 = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \frac{\text{Div}_3}{(1 + r_e)^3} + \cdots + \frac{\text{Div}_\infty}{(1 + r_e)^\infty} = \text{PV}(\text{Dividends})$$

- **Required return on equity** (r_e), the discount rate, set by the opportunity cost of investing in something with similar risk

Valuing Stocks, Dividend Discount Model

- Suppose you plan to buy a stock and own it forever
- What price should you pay for the stock today (P_0)?
- Stock value, the present value of the cash flows you get from owning it,

$$P_0 = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \frac{\text{Div}_3}{(1 + r_e)^3} + \cdots + \frac{\text{Div}_\infty}{(1 + r_e)^\infty} = \text{PV}(\text{Dividends})$$

- **Required return on equity** (r_e), the discount rate, set by the opportunity cost of investing in something with similar risk
- $\text{Div}_1, \text{Div}_2, \dots$ are *expected* dividends

Gordon Growth Model

- Special cases, using **Module 4**'s perpetuity formulas

Gordon Growth Model

- Special cases, using **Module 4**'s perpetuity formulas
- If dividends are expected to be constant,

$$P_0 = \frac{\text{Div}_1}{r_e}$$

Gordon Growth Model

- Special cases, using **Module 4**'s perpetuity formulas

- If dividends are expected to be constant,

$$P_0 = \frac{\text{Div}_1}{r_e}$$

- If dividends are expected to grow at a constant rate g ,

$$P_0 = \frac{\text{Div}_1}{r_e - g}$$

Gordon Growth Model

- Special cases, using **Module 4**'s perpetuity formulas

- If dividends are expected to be constant,

$$P_0 = \frac{\text{Div}_1}{r_e}$$

- If dividends are expected to grow at a constant rate g ,

$$P_0 = \frac{\text{Div}_1}{r_e - g}$$

- The latter formula is often called the “**Gordon growth model**”

Example, Gordon Growth Model

- 1. A firm will pay a \$3.00 dividend forever (the first dividend exactly a year from now). Its required return on equity is 12%. What will investors pay for the stock?
- 2. What if the firm's stock pays a \$3.00 dividend a year from now, and then the dividend grows at 8% forever?

Example, Gordon Growth Model

- 1. A firm will pay a \$3.00 dividend forever (the first dividend exactly a year from now). Its required return on equity is 12%. What will investors pay for the stock?
- 2. What if the firm's stock pays a \$3.00 dividend a year from now, and then the dividend grows at 8% forever?
- Solution, apply the perpetuity formulas,

$$1. \quad P_0 = \frac{\text{Div}_1}{r_e} = \frac{\$3.00}{0.12} = \$25$$

Example, Gordon Growth Model

- 1. A firm will pay a \$3.00 dividend forever (the first dividend exactly a year from now). Its required return on equity is 12%. What will investors pay for the stock?
- 2. What if the firm's stock pays a \$3.00 dividend a year from now, and then the dividend grows at 8% forever?
- Solution, apply the perpetuity formulas,

$$1. \quad P_0 = \frac{\text{Div}_1}{r_e} = \frac{\$3.00}{0.12} = \$25$$

$$2. \quad P_0 = \frac{\text{Div}_1}{r_e - g} = \frac{\$3.00}{0.12 - 0.08} = \$75$$

Valuing Stocks, One Period

- Suppose you plan to invest in a stock for only one year, and you want a fair expected return for the risk you take
- What price should you pay for the stock today (P_0)?

Valuing Stocks, One Period

- Suppose you plan to invest in a stock for only one year, and you want a fair expected return for the risk you take
- What price should you pay for the stock today (P_0)?
- Two sources of cash flow from owning the stock,
 1. Dividends paid during the year (Div_1), assume the dividend is paid at the end of the year
 2. The expected price from selling the shares at the end of the year ($E[P_1]$)

Valuing Stocks, One Period

- Suppose you plan to invest in a stock for only one year, and you want a fair expected return for the risk you take
- What price should you pay for the stock today (P_0)?
- Two sources of cash flow from owning the stock,
 1. Dividends paid during the year (Div_1), assume the dividend is paid at the end of the year
 2. The expected price from selling the shares at the end of the year ($E[P_1]$)
- Stock value today, the present value of these cash flows,

$$P_0 = \frac{\text{Div}_1 + E[P_1]}{1 + r_e}$$

Expected Return over One Period

- Rewrite the one-period equation as the expected return from the one-year investment,

$$P_0 = \frac{\text{Div}_1 + E[P_1]}{1 + r_e}$$

Expected Return over One Period

- Rewrite the one-period equation as the expected return from the one-year investment,

$$P_0 = \frac{\text{Div}_1 + E[P_1]}{1 + r_e}$$

- Solve for r_e ,

$$\text{Expected return } (r_e) = \frac{\text{Div}_1 + E[P_1] - P_0}{P_0} = \frac{\text{Div}_1}{P_0} + \frac{E[P_1] - P_0}{P_0}$$

Expected Return over One Period

- Rewrite the one-period equation as the expected return from the one-year investment,

$$P_0 = \frac{\text{Div}_1 + E[P_1]}{1 + r_e}$$

- Solve for r_e ,

$$\text{Expected return } (r_e) = \frac{\text{Div}_1 + E[P_1] - P_0}{P_0} = \frac{\text{Div}_1}{P_0} + \frac{E[P_1] - P_0}{P_0}$$

- That is, dividend yield plus expected capital appreciation (in %)

Example, Expected Return over One Year

- A firm plans to pay a dividend of \$5.00 this year
- Its shares sell for \$100 today. They are expected to sell for \$110 one year from now
- What is the expected return from investing in the firm's stock?

Example, Expected Return over One Year

- A firm plans to pay a dividend of \$5.00 this year
- Its shares sell for \$100 today. They are expected to sell for \$110 one year from now
- What is the expected return from investing in the firm's stock?
- Solution,

$$r_e = \frac{\text{Div}_1 + E[P_1] - P_0}{P_0} = \frac{\text{Div}_1}{P_0} + \frac{E[P_1] - P_0}{P_0}$$

Example, Expected Return over One Year

- A firm plans to pay a dividend of \$5.00 this year
- Its shares sell for \$100 today. They are expected to sell for \$110 one year from now
- What is the expected return from investing in the firm's stock?
- Solution,

$$\begin{aligned}r_e &= \frac{\text{Div}_1 + E[P_1] - P_0}{P_0} = \frac{\text{Div}_1}{P_0} + \frac{E[P_1] - P_0}{P_0} \\&= \frac{5}{100} + \frac{110 - 100}{100} = 5\% + 10\% = 15\%\end{aligned}$$

Limitations of the Dividend Discount Model

- Uncertain dividend forecasts
 - Dividends depend on the firm's future size, profitability, payout policy, leverage policy, and more

Limitations of the Dividend Discount Model

- Uncertain dividend forecasts
 - Dividends depend on the firm's future size, profitability, payout policy, leverage policy, and more
- How to value non-dividend-paying stocks?
 - If the firm is only temporarily not paying dividends, the dividend discount model still applies

Limitations of the Dividend Discount Model

- Uncertain dividend forecasts
 - Dividends depend on the firm's future size, profitability, payout policy, leverage policy, and more
- How to value non-dividend-paying stocks?
 - If the firm is only temporarily not paying dividends, the dividend discount model still applies
 - But future dividends are then probably even more uncertain!

Valuing Dividends with Non-Constant Growth

- Estimate the dividends year by year over a horizon of H years, then assume constant growth after year H ,

$$PV = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \cdots + \frac{\text{Div}_H}{(1 + r_e)^H} + \frac{P_H}{(1 + r_e)^H}$$

Valuing Dividends with Non-Constant Growth

- Estimate the dividends year by year over a horizon of H years, then assume constant growth after year H ,

$$PV = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \dots + \frac{\text{Div}_H}{(1 + r_e)^H} + \frac{P_H}{(1 + r_e)^H}$$

- Why? In practice, near-term dividends can be estimated carefully one year at a time, the distant future cannot

Valuing Dividends with Non-Constant Growth

- Estimate the dividends year by year over a horizon of H years, then assume constant growth after year H ,

$$PV = \frac{\text{Div}_1}{(1 + r_e)^1} + \frac{\text{Div}_2}{(1 + r_e)^2} + \cdots + \frac{\text{Div}_H}{(1 + r_e)^H} + \frac{P_H}{(1 + r_e)^H}$$

- Why? In practice, near-term dividends can be estimated carefully one year at a time, the distant future cannot
- You will learn more in “FINE 4100 Advanced Financial Management” and “FINE 4110 Investments in Equities”

Example, Valuing Non-Constant Growth

- A stock's expected dividends are in the table. Its required return on equity is 8.5%. Its dividends grow at 6% a year beginning in year 6. What is the value of the stock?

Year	1	2	3	4	5
Dividend	\$2.16	\$2.38	\$2.61	\$2.87	\$3.16

Example, Valuing Non-Constant Growth

- A stock's expected dividends are in the table. Its required return on equity is 8.5%. Its dividends grow at 6% a year beginning in year 6. What is the value of the stock?

Year	1	2	3	4	5
Dividend	\$2.16	\$2.38	\$2.61	\$2.87	\$3.16

- Solution, discount the five dividends, then the year-5 price of the constant-growth stream

Example, Valuing Non-Constant Growth

- A stock's expected dividends are in the table. Its required return on equity is 8.5%. Its dividends grow at 6% a year beginning in year 6. What is the value of the stock?

Year	1	2	3	4	5
Dividend	\$2.16	\$2.38	\$2.61	\$2.87	\$3.16

- Solution, discount the five dividends, then the year-5 price of the constant-growth stream

$$PV_{\text{Dividends 1-5}} = \frac{2.16}{1.085} + \frac{2.38}{1.085^2} + \frac{2.61}{1.085^3} + \frac{2.87}{1.085^4} + \frac{3.16}{1.085^5} = \$10.23$$

Example, Valuing Non-Constant Growth

- A stock's expected dividends are in the table. Its required return on equity is 8.5%. Its dividends grow at 6% a year beginning in year 6. What is the value of the stock?

Year	1	2	3	4	5
Dividend	\$2.16	\$2.38	\$2.61	\$2.87	\$3.16

- Solution, discount the five dividends, then the year-5 price of the constant-growth stream

$$PV_{\text{Dividends 1-5}} = \frac{2.16}{1.085} + \frac{2.38}{1.085^2} + \frac{2.61}{1.085^3} + \frac{2.87}{1.085^4} + \frac{3.16}{1.085^5} = \$10.23$$

$$PV_{\text{Price in year 5}} = \frac{P_5}{(1 + r_e)^5} = \frac{3.16 \times 1.06}{(0.085 - 0.06) \times 1.085^5} = \frac{133.98}{1.085^5} = \$89.11$$

Example, Valuing Non-Constant Growth

- A stock's expected dividends are in the table. Its required return on equity is 8.5%. Its dividends grow at 6% a year beginning in year 6. What is the value of the stock?

Year	1	2	3	4	5
Dividend	\$2.16	\$2.38	\$2.61	\$2.87	\$3.16

- Solution, discount the five dividends, then the year-5 price of the constant-growth stream

$$PV_{\text{Dividends 1-5}} = \frac{2.16}{1.085} + \frac{2.38}{1.085^2} + \frac{2.61}{1.085^3} + \frac{2.87}{1.085^4} + \frac{3.16}{1.085^5} = \$10.23$$

$$PV_{\text{Price in year 5}} = \frac{P_5}{(1 + r_e)^5} = \frac{3.16 \times 1.06}{(0.085 - 0.06) \times 1.085^5} = \frac{133.98}{1.085^5} = \$89.11$$

$$PV_{\text{Stock}} = 10.23 + 89.11 = \$99.33$$

Practice Problem

- Consider the following three stocks,
 - a. Stock A is expected to pay a dividend of \$10 a share forever
 - b. Stock B is expected to pay a dividend of \$5 next year. Thereafter, dividend growth is expected to be 4% a year forever
 - c. Stock C is expected to pay a dividend of \$5 next year. Thereafter, dividend growth is expected to be 20% a year for 5 years (years 2 through 6) and zero thereafter
- Each stock's required return on equity is 10%. Which stock is the most valuable? What if the required return on equity is 7%?

Practice Problem

- Solution, at a 10% required return on equity,

10% capitalization rate:

$$P_A = \text{DIV}_1 / r = \$10 / 0.1 = \$100$$

$$P_B = \text{DIV}_1 / (r - g) = \$5 / (0.1 - 0.04) = \$83.33$$

$$P_C = \frac{\text{DIV}_1}{1+r} + \frac{\text{DIV}_1 \times (1+g)}{(1+r)^2} + \frac{\text{DIV}_1 \times (1+g)^2}{(1+r)^3} + \frac{\text{DIV}_1 \times (1+g)^3}{(1+r)^4} + \frac{\text{DIV}_1 \times (1+g)^4}{(1+r)^5} +$$

$$\frac{\text{DIV}_1 \times (1+g)^5}{(1+r)^6} + \frac{(\text{DIV}_1 \times [1+g]^5 \times [1+g_2]) / r}{(1+r)^6}$$

$$P_C = \$5 / 1.1 + (\$5 \times 1.2) / 1.1^2 + (\$5 \times 1.2^2) / 1.1^3 + (\$5 \times 1.2^3) / 1.1^4 \\ + (\$5 \times 1.2^4) / 1.1^5 + (\$5 \times 1.2^5) / 1.1^6 + ([\$5 \times 1.2^5 \times \{1 + 0\}] / 0.1) / 1.1^6$$

$$P_C = \$104.51$$

At a 10% capitalization rate, Stock C has the largest present value.

Practice Problem

- Solution, at a 10% required return on equity,

10% capitalization rate:

$$P_A = \text{DIV}_1 / r = \$10 / 0.1 = \$100$$

$$P_B = \text{DIV}_1 / (r - g) = \$5 / (0.1 - 0.04) = \$83.33$$

$$\begin{aligned} P_C &= \frac{\text{DIV}_1}{1+r} + \frac{\text{DIV}_1 \times (1+g)}{(1+r)^2} + \frac{\text{DIV}_1 \times (1+g)^2}{(1+r)^3} + \frac{\text{DIV}_1 \times (1+g)^3}{(1+r)^4} + \frac{\text{DIV}_1 \times (1+g)^4}{(1+r)^5} + \\ &\quad \frac{\text{DIV}_1 \times (1+g)^5}{(1+r)^6} + \frac{(\text{DIV}_1 \times [1+g]^5 \times [1+g_2]) / r}{(1+r)^6} \\ P_C &= \$5 / 1.1 + (\$5 \times 1.2) / 1.1^2 + (\$5 \times 1.2^2) / 1.1^3 + (\$5 \times 1.2^3) / 1.1^4 \\ &\quad + (\$5 \times 1.2^4) / 1.1^5 + (\$5 \times 1.2^5) / 1.1^6 + ([\$5 \times 1.2^5 \times \{1 + 0\}] / 0.1) / 1.1^6 \\ P_C &= \$104.51 \end{aligned}$$

At a 10% capitalization rate, Stock C has the largest present value.

- At a 7% required return on equity, $P_A = \$142.86$, $P_B = \$166.67$, and $P_C = \$156.50$, so Stock B is the most valuable

Valuing Stocks, Comparables

- What is one of the most common approaches to valuing real estate?

Valuing Stocks, Comparables

- What is one of the most common approaches to valuing real estate?
- Comparables (“comps”)!

Valuing Stocks, Comparables

- What is one of the most common approaches to valuing real estate?
- Comparables (“comps”)!
- E.g., a 2,000 sqft house in New Orleans sells for \$400,000. What is your estimate for a neighboring house of 2,500 sqft?

Valuing Stocks, Comparables

- Real estate comparables use price per square foot. For companies, the most common metric is P/E (price to EPS)

Valuing Stocks, Comparables

- Real estate comparables use price per square foot. For companies, the most common metric is P/E (price to EPS)
- E.g., Company A creates twice the cash flow per share of Company B, and Company B trades at \$10. What price should Company A's shares trade at?

Valuing Stocks, Comparables

- Real estate comparables use price per square foot. For companies, the most common metric is P/E (price to EPS)
- E.g., Company A creates twice the cash flow per share of Company B, and Company B trades at \$10. What price should Company A's shares trade at?
- Underlying idea, we can “scale” prices!

Valuing Stocks by Comparables, Four Steps

- Suppose we want to value Company X

Valuing Stocks by Comparables, Four Steps

- Suppose we want to value Company X
- 1. Start with a sample of “comparable” companies (A, B, C, ...) whose business characteristics are similar to Company X

Valuing Stocks by Comparables, Four Steps

- Suppose we want to value Company X
- 1. Start with a sample of “comparable” companies (A, B, C, ...) whose business characteristics are similar to Company X
- 2. Pick a ratio (“multiple”) with the stock price or equity value in the numerator, e.g., P/E or market-to-book

Valuing Stocks by Comparables, Four Steps

- Suppose we want to value Company X
 1. Start with a sample of “comparable” companies (A, B, C, ...) whose business characteristics are similar to Company X
 2. Pick a ratio (“multiple”) with the stock price or equity value in the numerator, e.g., P/E or market-to-book
 3. Calculate the average of that multiple for the comparables A, B, C, ...

Valuing Stocks by Comparables, Four Steps

- Suppose we want to value Company X
- 1. Start with a sample of “comparable” companies (A, B, C, ...) whose business characteristics are similar to Company X
- 2. Pick a ratio (“multiple”) with the stock price or equity value in the numerator, e.g., P/E or market-to-book
- 3. Calculate the average of that multiple for the comparables A, B, C, ...
- 4. Back out Company X’s implied value (the numerator) as the average multiple times Company X’s denominator

$$P_X = (\text{Average P/E of comparables}) \times \text{EPS}_X$$

Example, Valuation by Comparables

- Furniture manufacturer Herman Miller has earnings per share of \$1.38
- The average P/E of comparable furniture stocks is 21.3
- What is your estimate of the value of Herman Miller's stock using the comparables method?

Example, Valuation by Comparables

- Furniture manufacturer Herman Miller has earnings per share of \$1.38
- The average P/E of comparable furniture stocks is 21.3
- What is your estimate of the value of Herman Miller's stock using the comparables method?
- Solution, multiply Herman Miller's EPS by the P/E of comparable firms,
$$P = (\text{Average P/E of comparables}) \times \text{EPS} = 21.3 \times 1.38 = \$29.39$$

Example, Valuation by Comparables

- Citigroup's book value of equity is \$23 billion. It has 1.2 billion shares outstanding. The average market-to-book ratio of similar banks is 1.6
- Using the comparables method, what is the estimate of Citigroup's stock price?

Example, Valuation by Comparables

- Citigroup's book value of equity is \$23 billion. It has 1.2 billion shares outstanding. The average market-to-book ratio of similar banks is 1.6
- Using the comparables method, what is the estimate of Citigroup's stock price?
- Solution, the comparables imply an estimate of Citigroup's market capitalization,
 $M_{\text{Citigroup}} = (\text{Average M/B of comparables}) \times B_{\text{Citigroup}} = 1.6 \times 23 = \36.8 billion

Example, Valuation by Comparables

- Citigroup's book value of equity is \$23 billion. It has 1.2 billion shares outstanding. The average market-to-book ratio of similar banks is 1.6
- Using the comparables method, what is the estimate of Citigroup's stock price?
- Solution, the comparables imply an estimate of Citigroup's market capitalization,
 $M_{\text{Citigroup}} = (\text{Average M/B of comparables}) \times B_{\text{Citigroup}} = 1.6 \times 23 = \36.8 billion

- The question asks for the share price, not the market cap,

$$P = \frac{\text{Market cap}}{\text{Number of shares outstanding}} = \frac{36.8 \text{ billion}}{1.2 \text{ billion}} = \$30.67$$

Limitations of the Comparables Method

- The estimate is sensitive to the choice of multiple
 - E.g., P/E, market-to-book, or something else

Limitations of the Comparables Method

- The estimate is sensitive to the choice of multiple
 - E.g., P/E, market-to-book, or something else
- Good comparable companies are hard to find
 - Firms differ in managerial talent, manufacturing processes, patents, customer base
 - Matters if the differences affect future cash flow growth or the risks the firms face

Limitations of the Comparables Method

- The estimate is sensitive to the choice of multiple
 - E.g., P/E, market-to-book, or something else
- Good comparable companies are hard to find
 - Firms differ in managerial talent, manufacturing processes, patents, customer base
 - Matters if the differences affect future cash flow growth or the risks the firms face
- Assumes the comparables are priced correctly
 - Cannot tell if an entire industry is overvalued or undervalued
 - E.g., no help in valuing tech firms during the late-1990s “dot-com bubble”

Buying and Selling Stock on the Stock Market

- Current stock price

FedEx Corporation (FDX)

NYSE - Nasdaq Real Time Price. Currency in USD

☆ [Add to watchlist](#)

239.98 -9.04 (-3.63%)

As of 1:02PM EDT. Market open.

Summary

[Chart](#)

[Conversations](#)

[Statistics](#)

[Profile](#)

[Final](#)

Previous Close	249.02	Market Cap	64.318B
Open	246.00	Beta	1.57
Bid	239.98 x 100	PE Ratio (TTM)	22.11
Ask	240.10 x 100	EPS (TTM)	10.86
Day's Range	236.00 - 246.00	Earnings Date	Mar 19, 2018 - Mar 23, 2018
52 Week Range	182.89 - 274.66	Forward Dividend & Yield	2.00 (0.80%)
Volume	2,359,358	Ex-Dividend Date	N/A
Avg. Volume	1,866,076	1y Target Est	284.69

Buying and Selling Stock on the Stock Market

- Current stock price
- **Bid**, the price at which investors are willing to buy shares
- **Ask**, the price at which current investors are willing to sell shares

FedEx Corporation (FDX)

NYSE - Nasdaq Real Time Price. Currency in USD

☆ Add to watchlist

239.98 -9.04 (-3.63%)

As of 1:02PM EDT. Market open.

Summary				Chart	Conversations	Statistics	Profile	Final
Previous Close	249.02	Market Cap	64.318B					
Open	246.00	Beta	1.57					
Bid	239.98 x 100	PE Ratio (TTM)	22.11					
Ask	240.10 x 100	EPS (TTM)	10.86					
Day's Range	236.00 - 246.00	Earnings Date	Mar 19, 2018 - Mar 23, 2018					
52 Week Range	182.89 - 274.66	Forward Dividend & Yield	2.00 (0.80%)					
Volume	2,359,358	Ex-Dividend Date	N/A					
Avg. Volume	1,866,076	1y Target Est	284.69					

Buying and Selling Stock on the Stock Market

- Current stock price
- **Bid**, the price at which investors are willing to buy shares
- **Ask**, the price at which current investors are willing to sell shares

FedEx Corporation (FDX)

☆ Add to watchlist

NYSE - Nasdaq Real Time Price. Currency in USD

239.98 -9.04 (-3.63%)

As of 1:02PM EDT. Market open.

Summary				Chart	Conversations	Statistics	Profile	Final
Previous Close	249.02	Market Cap	64.318B					
Open	246.00	Beta	1.57					
Bid	239.98 x 100	PE Ratio (TTM)	22.11					
Ask	240.10 x 100	EPS (TTM)	10.86					
Day's Range	236.00 - 246.00	Earnings Date	Mar 19, 2018 - Mar 23, 2018					
52 Week Range	182.89 - 274.66	Forward Dividend & Yield	2.00 (0.80%)					
Volume	2,359,358	Ex-Dividend Date	N/A					
Avg. Volume	1,866,076	1y Target Est	284.69					

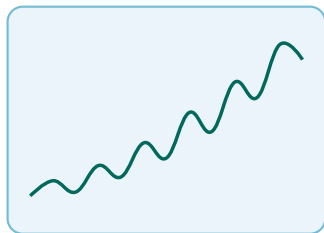
- Market capitalization
- **Beta**, a measure of risk (how much the stock moves with the overall market)
- Price/earnings
- EPS
- Dividend and dividend yield

Can You Beat the Stock Market?

- It's really hard!

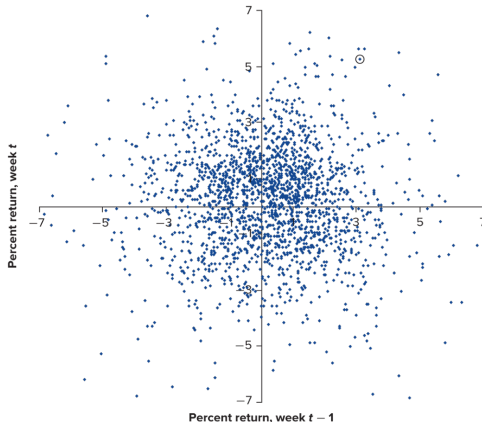
Can You Beat the Stock Market?

- It's really hard!
- **Technical analysis**, investors who attempt to forecast stock prices by watching the fluctuations in historical prices (“wiggle watchers”)



Discuss, Where Is the Pattern?

- Scatter plot of the NYSE Composite Index return in one week against its return in the previous week



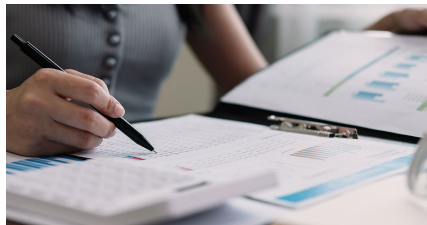
Can You Beat the Stock Market?

- **Fundamental analysis**, investors who attempt to find mispriced securities by analyzing fundamental information, such as accounting data and business prospects

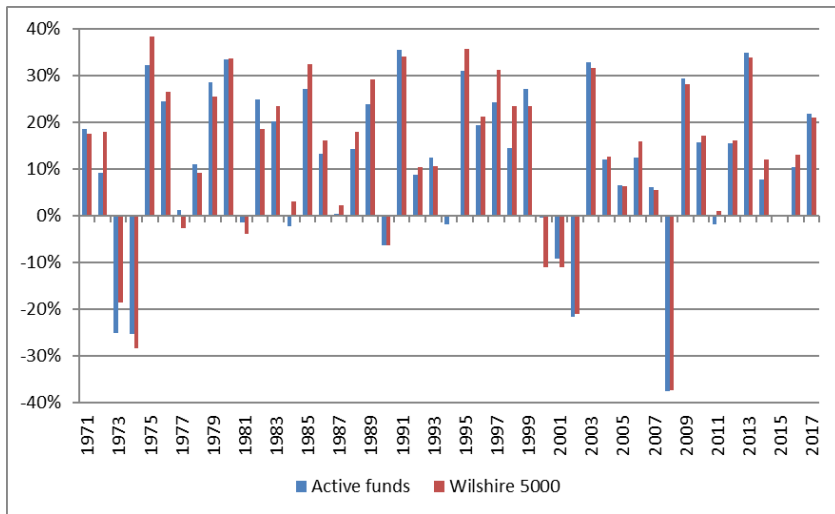


Can You Beat the Stock Market?

- **Fundamental analysis**, investors who attempt to find mispriced securities by analyzing fundamental information, such as accounting data and business prospects
- Also really hard!



Professional Fund Managers Don't Generally Beat the Index!



Information in Stock Prices

- Why is it so hard for professionals to beat a passive indexing strategy?

Information in Stock Prices

- Why is it so hard for professionals to beat a passive indexing strategy?
- For a publicly traded firm, the **stock price** already provides very accurate information about the true value of the shares

Information in Stock Prices

- Why is it so hard for professionals to beat a passive indexing strategy?
- For a publicly traded firm, the **stock price** already provides very accurate information about the true value of the shares
- **Efficient markets hypothesis**, “securities are fairly priced, given all information that is available to investors”

Information in Stock Prices

- Why is it so hard for professionals to beat a passive indexing strategy?
- For a publicly traded firm, the **stock price** already provides very accurate information about the true value of the shares
- **Efficient markets hypothesis**, “securities are fairly priced, given all information that is available to investors”
- What **type** of information is captured in the stock price?
 1. All historical price information? (“weak” efficiency)
 2. All public, easily available information? (“semi-strong” efficiency)
 3. Also all private or difficult-to-interpret information? (“strong” efficiency)

What Does This Mean for Investors?

Pros 👍

- If stocks are fairly priced, even an investor who **knows nothing** can buy **any** stock and expect it to fairly compensate them for the risk they take

Cons 👎

- To *consistently* make money above the *fair* (*risk-compensating*) return by betting on publicly traded shares, an investor needs a very distinct competitive advantage
- E.g., expertise or access to information that only a few people can figure out

Recap of Part 2

- Dividend discount model, the stock value is the present value of its expected dividends at the required return on equity r_e

Recap of Part 2

- Dividend discount model, the stock value is the present value of its expected dividends at the required return on equity r_e
- Constant dividends, $P_0 = \text{Div}_1 / r_e$, constant growth (Gordon growth model), $P_0 = \text{Div}_1 / (r_e - g)$

Recap of Part 2

- Dividend discount model, the stock value is the present value of its expected dividends at the required return on equity r_e
- Constant dividends, $P_0 = \text{Div}_1 / r_e$, constant growth (Gordon growth model), $P_0 = \text{Div}_1 / (r_e - g)$
- One period, expected return = dividend yield + expected capital appreciation

Recap of Part 2

- Dividend discount model, the stock value is the present value of its expected dividends at the required return on equity r_e
- Constant dividends, $P_0 = \text{Div}_1 / r_e$, constant growth (Gordon growth model), $P_0 = \text{Div}_1 / (r_e - g)$
- One period, expected return = dividend yield + expected capital appreciation
- Comparables, scale a peer multiple (P/E, market-to-book) to the firm being valued

Recap of Part 2

- Dividend discount model, the stock value is the present value of its expected dividends at the required return on equity r_e
- Constant dividends, $P_0 = \text{Div}_1 / r_e$, constant growth (Gordon growth model), $P_0 = \text{Div}_1 / (r_e - g)$
- One period, expected return = dividend yield + expected capital appreciation
- Comparables, scale a peer multiple (P/E, market-to-book) to the firm being valued
- Beating the market is really hard, prices already reflect available information (the efficient markets hypothesis)

Key Takeaways

- Bonds and stocks, the present value of their expected cash flows (coupons and face value, dividends)

Key Takeaways

- Bonds and stocks, the present value of their expected cash flows (coupons and face value, dividends)
- Bonds, the yield to maturity discounts the promised payments, the required rate of return r the expected ones

Key Takeaways

- Bonds and stocks, the present value of their expected cash flows (coupons and face value, dividends)
- Bonds, the yield to maturity discounts the promised payments, the required rate of return r the expected ones
- Interest rates and bond prices move in opposite directions

Key Takeaways

- Bonds and stocks, the present value of their expected cash flows (coupons and face value, dividends)
- Bonds, the yield to maturity discounts the promised payments, the required rate of return r the expected ones
- Interest rates and bond prices move in opposite directions
- Stocks, the dividend discount model or comparables, and efficient markets make the index hard to beat

Thank you!

Please let me know if you have any questions.